

18. UI Wireframes and Screen Concepts

Annotated wireframes and concept screens for the first design baseline

Program	QuantPipeline - private modular portfolio decision platform
Document purpose	Annotated wireframes and concept screens for the first design baseline
Status	Working baseline for design and delivery

This document translates the IA and UX rules into concrete screen concepts. The wireframes are intentionally mid-fidelity: they define hierarchy, zones, and behavioral responsibilities without pretending that the final visual design is settled.

1. Wireframing objectives

- Lock screen hierarchy before visual polish begins.
- Verify that the product can expose recommendation, evidence, risk, and action in one coherent canvas.
- Ensure that mobile and operator surfaces inherit the same mental model.

2. Global layout conventions

Pattern	Definition	Why it matters
Three-zone desktop	Left navigation, center canvas, right context pane	gives stable orientation and separates detail from actions
Hero-first content stack	top summary before deep evidence	matches the decision rhythm
Action persistence	primary action bar visible near recommendation context	reduces accidental context loss
Mobile bottom-sheet context	right-rail content becomes layered bottom sheet	preserves explainability on narrow screens

3. Main decision workspace concept

The main decision workspace is the most important screen in the product. The recommendation hero, risk summary, and disagreement block are above the fold. Evidence narrative and challenge controls occupy the center. The right context pane is never decorative; it is a working surface for risk, confidence, provenance, and action.

Screen concept: main decision workspace

Desktop wireframe with persistent context pane

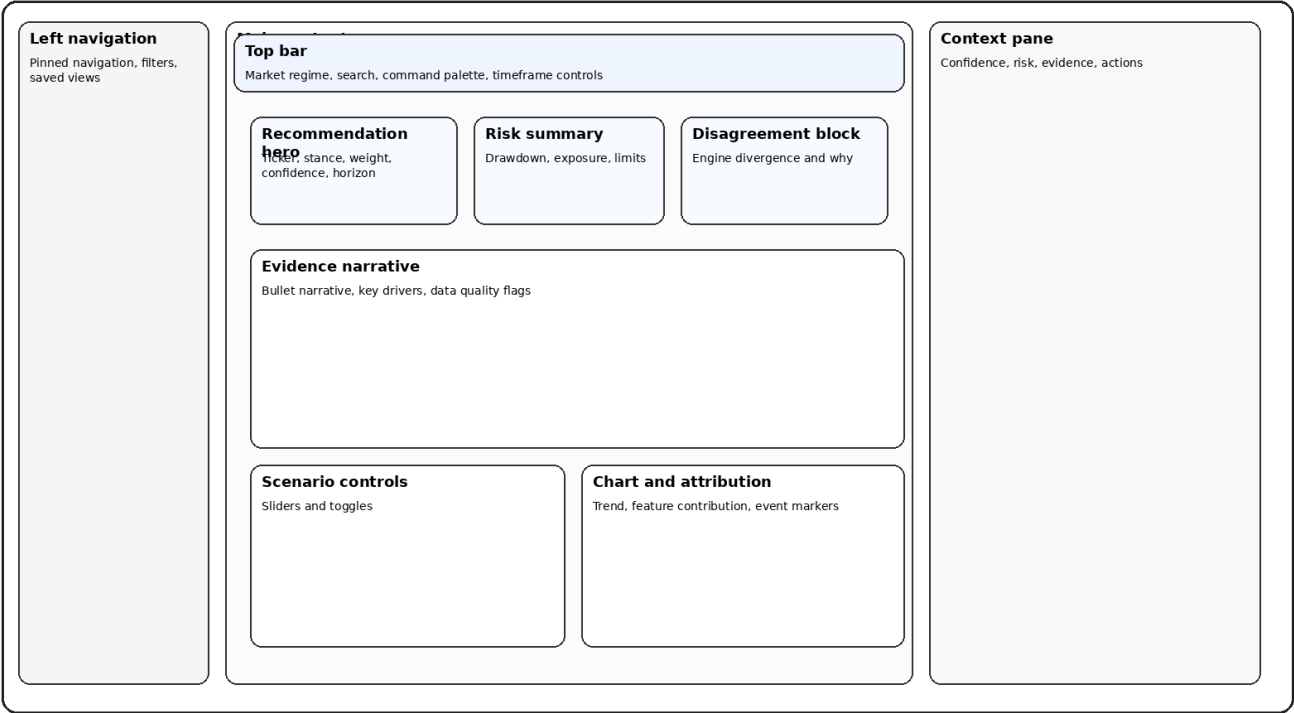


Figure 18-1. Main decision workspace wireframe.

Region	Must show	Interaction notes
Left navigation	major sections, saved views, active route	collapsed state still exposes active location
Top bar	scope, timeframe, search, mode	must remain visible during internal workspace scroll
Recommendation hero	stance, weight, horizon, confidence	one-click access to notes and publication state
Evidence narrative	why now, key drivers, data caveats	supports inline expansion, not page replacement
Scenario controls	assumption edits and sensitivity toggles	show predicted delta before apply
Context pane	risk, disagreement, provenance, actions	tabbed only if density becomes unmanageable

4. Engine comparison concept

Comparison should not feel like a separate product. It answers one question: how do multiple analytical engines differ, and what synthesis should the user trust?

Screen concept: engine comparison workspace

Structured comparison before synthesis

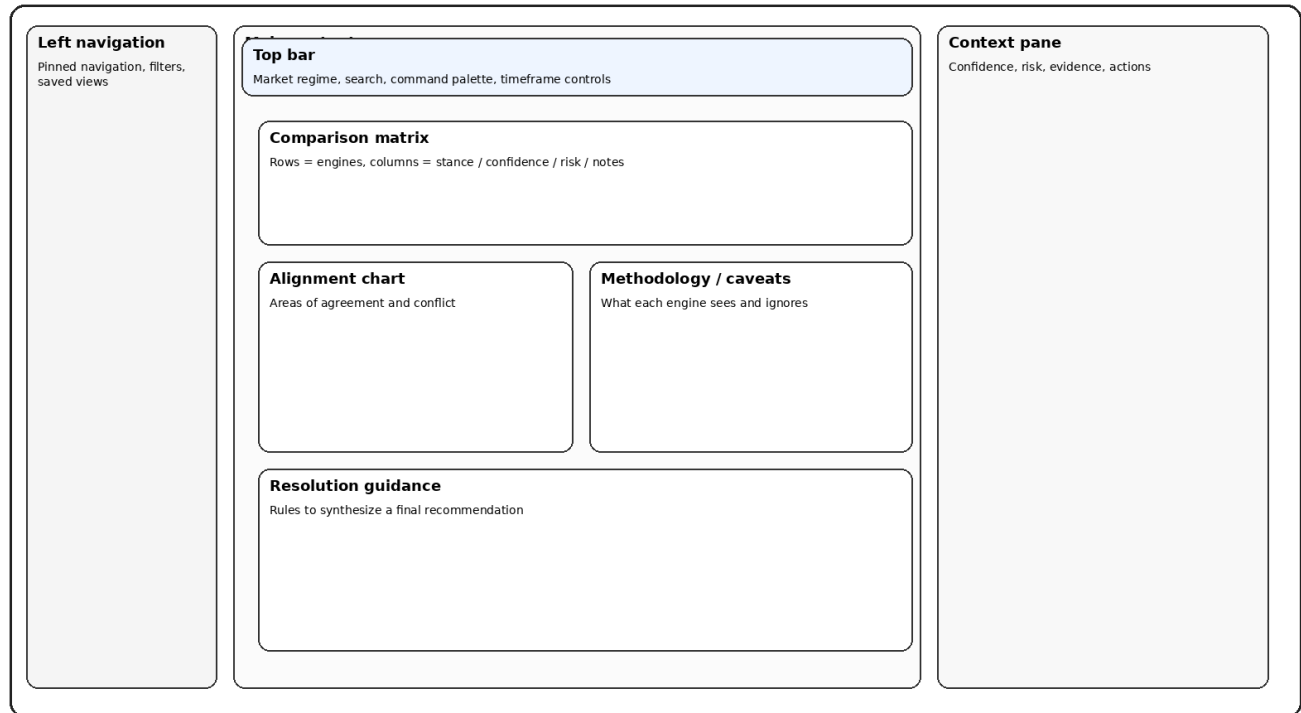


Figure 18-2. Engine comparison workspace wireframe.

- Use a structured matrix for direct comparison across stance, confidence, risk, and notes.
- Supplement the matrix with one visual explaining where alignment is high or low.
- Finish with a synthesis panel that guides the user back into the decision workspace.

5. Mobile decision concept

Mobile supports triage, monitoring, and lightweight challenge. It should surface a concise recommendation card first, then stack evidence and risk in digestible modules.

Screen concept: mobile decision workspace

Portable summary-first experience

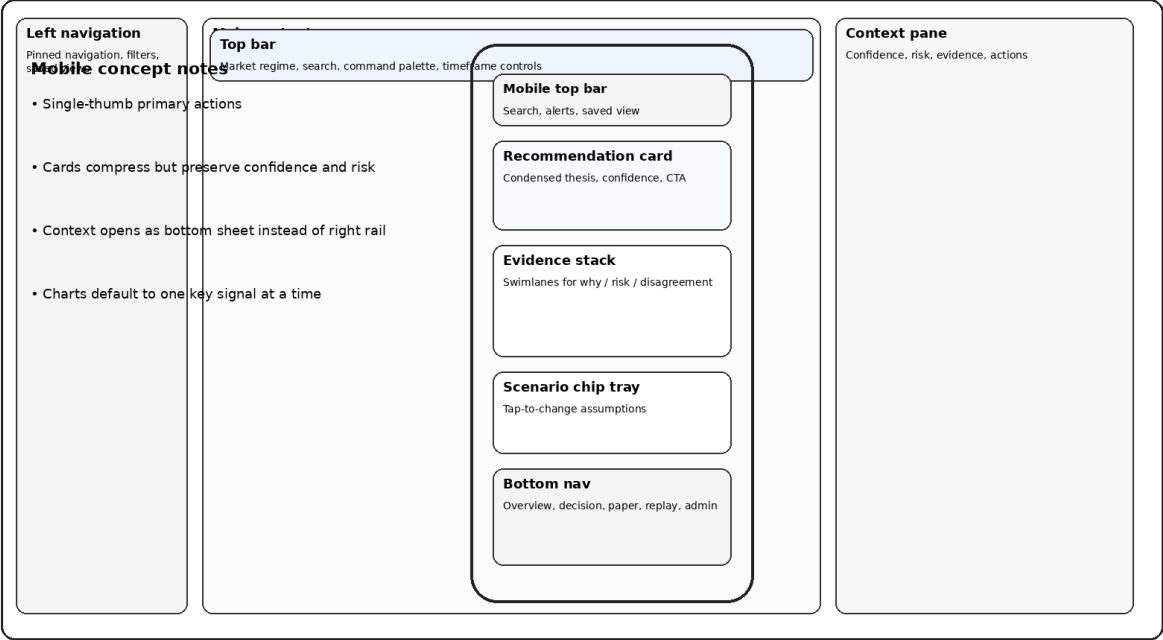


Figure 18-3. Mobile decision workspace wireframe.

Mobile rule	Description
Card-first hierarchy	The recommendation card precedes charts and deep notes
Bottom navigation	Core product sections remain reachable with one thumb
Context as sheet	Detailed evidence, risk, and notes open as bottom sheets
Progressive charts	Only one primary chart per viewport by default

6. Admin and ops concept

The operator view privileges queue state, health, policy, and investigation speed. It still shares the same shell and object model so that admin work remains connected to what the end user sees.

Screen concept: admin and ops workspace

Operator-first workspace for publication and incident handling

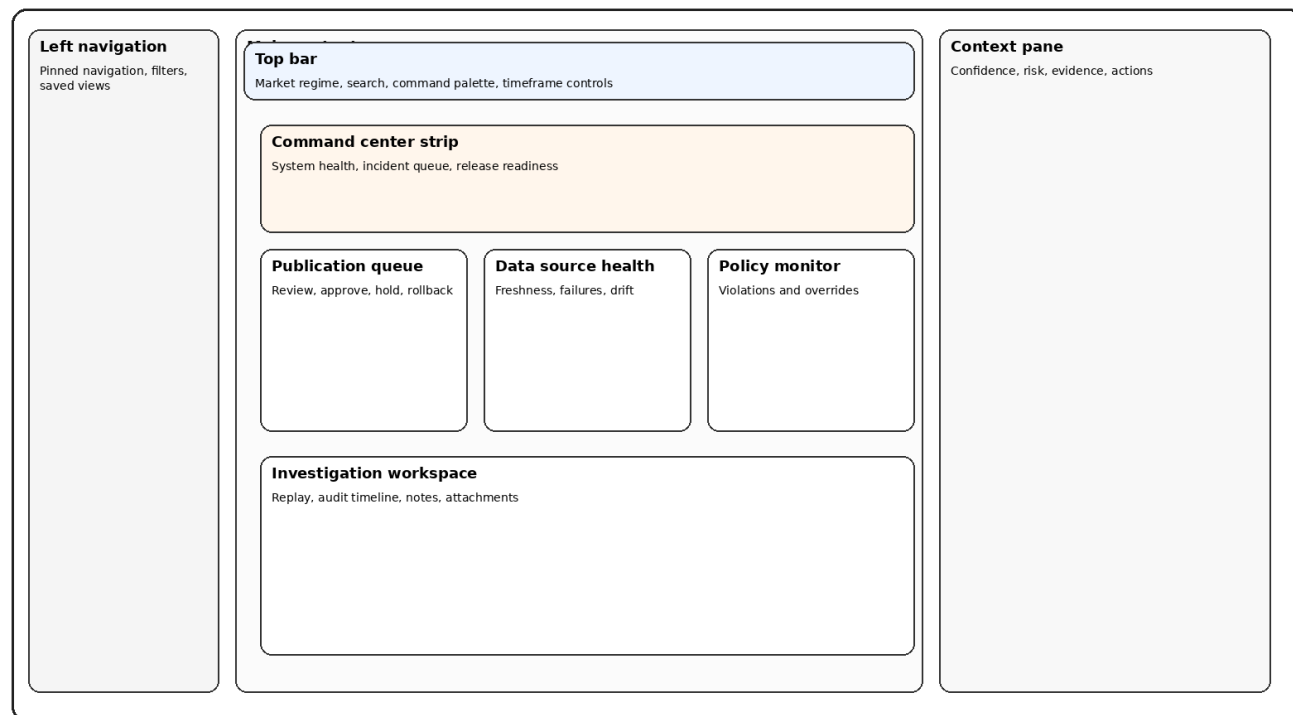


Figure 18-4. Admin and ops workspace wireframe.

Admin block	Responsibility
Command center strip	At-a-glance health and release readiness
Publication queue	Approval, hold, rollback, and reason capture
Data source health	freshness, reliability, and drift awareness
Policy monitor	violations, overrides, and approval logic
Investigation workspace	replay-driven investigation and audit notes

7. Secondary concepts to design next

- Backtest lab with assumptions panel and result comparison
- Paper portfolio review with proposal staging and drift alerts
- Replay investigator timeline with artifact side panel
- Universe management with screening, tagging, and saved cohorts

8. Wireframe acceptance checklist

Question	Pass condition
Is the recommendation visible without scrolling on core screens?	yes
Can the user challenge assumptions without losing the baseline?	yes

Do admin screens remain visually related to user screens?	yes
Does mobile preserve trust signals instead of hiding them?	yes

9. Engineering handoff annotations

The wireframes should be read as layout and behavior contracts rather than visual suggestions only. During implementation the following annotations must stay attached to each core screen concept.

Annotation type	Examples	Why it must exist
Persistent context	selected recommendation, scenario values, current horizon	prevents accidental loss of state across panels and routes
Deferred loading	comparison details, long histories, extended evidence	keeps first render fast while preserving depth
Role-aware actions	publish, hold, rollback, policy override	avoids shipping user-only assumptions into operator routes
Fallback rendering	missing chart data, unavailable attribution, stale inputs	prevents blank analytical surfaces

10. Responsive adaptation matrix

Screen concept	Desktop priority	Tablet adaptation	Mobile adaptation
Decision workspace	three-zone layout	context rail may collapse	context becomes bottom sheet
Comparison workspace	matrix plus synthesis panel	matrix can scroll horizontally	accordion rows with synthesis summary
Admin workspace	parallel modules visible	secondary modules stack lower	critical queue and incident modules only

11. Screen concept sign-off criteria

- A designer and engineer should be able to agree on component boundaries from the wireframes alone.
- A product reviewer should be able to point to the primary action on every concept screen without guessing.
- Every concept should make degraded and empty states plausible, even if not all are drawn directly on the wireframe.